

GtelPay Core — Design Sign-Off

Field	Value
Document	Design & verified-core sign-off
Component	core.accounting + core.wallet (+ core.foundation, app-orchestration)
Domain	Fiat payments (VND)
Status	Draft — awaiting approval
Prepared for	Architecture / Engineering / Finance / Product approval

Purpose. This document records approval that the GtelPay Core **design** and its **implemented core** are sound, scoped, and verified — and may proceed to the next phase. Section 6 lists open items so that approval is **informed**, not blind.

1. Scope being signed off

In scope - core.accounting — immutable double-entry ledger (chart of accounts, journals, transit, EOD). - core.wallet — per-member spendable balance (available / frozen), append-only movements. - core.foundation — shared lib (envelope, errors, money utils). - app-orchestration — saga sequencer across the two domains. - Currency: **VND (fiat) only**.

Explicitly out of scope (not part of this sign-off) - Crypto / hot-cold custody / C2C-OTC escrow / multi-currency. - The reference codebase (00_framework, CoinExchange/BIZZAN) — technical reference only.

2. What is delivered

Area	Delivered
Design decisions	41 ADRs , each with acceptance criteria (AC) + test cases (TC)
Behaviour	Process flows (spec/processes.md), foundation fund flows
Contracts	OpenAPI + AsyncAPI (spec/contracts/) as the wire source of truth
Conformance	150+ acceptance scenarios (Given/When/Then)
Implementation	core.foundation · core.wallet · core.accounting · app-orchestration (Java / Spring Boot 3) — building
Automated tests	38 tests passing (mvn test, BUILD SUCCESS)

3. Correctness evidence (the safeguards are real and tested)

Invariant	Guarantee	Verified by
Idempotency (under lock)	A retry never double-applies / double-spends	concurrentSameTriple_appliesOnceAndReplaysLoser, duplicateBusinessRef_*
Double-entry balanced	Every journal: debits = credits, else rejected	unbalancedLines_rejectsPost
Immutable ledger	A POSTED line cannot be edited	postedLine_tamperIsIgnored_immutable
Transit net-zero (all use cases)	No money stranded mid-flow at POST	balancedButStrandedTransit_rejectsPost, payment...transit3500Zero
Period close	No posting into a CLOSED/LOCKED period	closedPeriod_rejectsPost
Outflow holds funds	Settle debits frozen , not available (no double-spend)	withdrawFreezeThenSettle, ibftFreezeThenSettle_deductsFrozenNotAvailable
Wallet snapshot + log together	No balance change without a movement row, same TX	creditThenDebit_balanceMatches, balance integration tests

4. Recently closed gaps (A1–A4)

#	Gap	Fix	Commit
A1	Transit net-zero enforced only for deposit	Enforced for every use case at post time	a362798
A2	IBFT settle could debit available → double-spend	IBFT_SETTLE deducts from frozen	a362798
A3	Ledger immutability was convention-only	@Immutable hard ORM guard	66fd72e
A4	Period close (ADR-023) not implemented	coa_period + post-time period guard	66fd72e
—	Proof tests for A1–A4	red/green tests added	ca06314, c28ce9e

5. Quality gates

- `mvn test` → **BUILD SUCCESS, 38 tests pass** (foundation 9 · wallet 16 · accounting 13).
 - SQL ledger-invariant suite gates CI (ADR-031): balanced journals, transit net-zero.
 - Contracts (OpenAPI/AsyncAPI) validated; pocket/error schemas resolve.
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6. Open items & known limitations (approval is informed)

Sign-off does not claim these are done. They are accepted as known and deferred/tracked:

1. **Multi-pocket (ADR-040)** is designed, spec'd, and contract-drafted, but **not yet implemented** in `core.wallet` code.
 2. **Period management is minimal** — a period can be marked CLOSED/LOCKED, but there is no automated period-end / close workflow yet.
 3. **Operational surface** — RabbitMQ workers, the outbox relay, and reconciliation jobs require ops capacity to run and monitor.
 4. **Eventual consistency** between wallet and accounting — a lag window exists; W5 reconciliation is **report-only** (drift requires manual follow-up, never auto-corrected).
 5. **Not load-tested** — NFR figures (RPS, P95) are **targets**, not measured results.
 6. **Documentation spans** `design/`, `design-v2/`, `spec/`, `specs/` — consolidation is pending.
 7. **Wallet tx-type catalog** in code is a subset of the design catalog (PARTNER/PAYROLL/MERCHANT_SETTLE types not yet added).
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7. Sign-off

By signing, the approver confirms they have reviewed Sections 1–6 and approve the design and verified core to proceed, subject to any conditions noted.

Role	Name	Decision (Approve / Approve-with-conditions / Reject)	Conditions / Notes	Date	Signature
Architecture					
Engineering Lead					
Accounting / Finance owner					
Product owner					
Compliance / Audit (if applicable)					

Overall decision: ☐ Approved ☐ Approved with conditions ☐ Rejected

8. References (backing)

adr/ (41 ADRs + AC/TC) · spec/ (foundation, processes, contracts) · design-
v2/acceptance.md (150+ scenarios) · platform/ (running modules + tests) · DESIGN-
BRIEF.md / DESIGN-DECK.md (summary + slides).